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Exit without choice: interpretable machine learning unlocks the structural drivers of smallholder dispossession in Pakistan

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Abstract

In Pakistan's evolving agricultural landscape, smallholder farmers are increasingly exiting farming, not by choice but under structural pressure. This study investigates the underlying associations using an integrated machine learning and econometric approach, based on novel data from 500 current and former farmers. We employ CatBoost, a high-performance gradient boosting algorithm designed to handle categorical data efficiently, in combination with recursive feature elimination and cross-validation to identify seven key features of exit. CatBoost demonstrates robust performance on both imbalanced and balanced datasets. SHapley Additive Explanations (SHAP) analysis reveals that reliance on full credit, high debt, distant markets, and natural hazards significantly increase exit risk. Conversely, larger landholdings, non-farming income, and livestock ownership serve as protective factors. Logistic regression confirms these associations, with marginal effects indicating that reliance on credit rather than cash is associated with a 9.5% higher probability of exit, and a 22.6% increase for high debt. Market distance is associated with an 18.3% increase, whereas each additional unit of land size reduces exit risk by 15.5%. Subgroup analysis reveals exit rates of 72.6% among credit-dependent farmers and 56.0% among those with less than five acres, compared to none among large landowners. By framing exit as structural exclusion, this study offers a scalable, explainable AI tool for inclusive, faith-sensitive rural reform aligned with food security, poverty reduction, and multiple Sustainable Development Goals.

Keywords: Agricultural exit, Smallholder farmers, Machine learning, Debt traps, Religious financial constraints

Introduction

Agriculture remains the cornerstone of developing economies, particularly in Pakistan, where it sustains a significant portion of the labor force and serves as a vital pillar for food security, poverty reduction, and rural livelihoods (Ali et al. 2024). Yet a troubling paradox persists: Despite increased agricultural output, rural poverty continues to deepen, especially among smallholder farmers. According to official statistics, agriculture grew by 6.25% and crop production by 11.03% during 2023–24 (Finance Division, 2024). However, the national poverty rate surged to 25.3% in 2024, marking a 7%

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increase from the previous year and pushing an additional 13 million people below the poverty line (Oscar Barriga-Cabanillas, 2024). Once central to agricultural progress, many smallholder farmers are now leaving farming altogether. This contradiction raises urgent questions: Why is poverty worsening amid growth? What systemic barriers are driving smallholder farmers away from agriculture? And how can inclusive, sustainable rural development be achieved? These questions are serious for meeting the Sustainable Development Goals (SDGs) on poverty eradication, food security, and equitable rural growth.

To clarify the study's purpose and conceptual boundaries, this study is primarily diagnostic and exploratory. It seeks to identify and explain the key socioeconomic, institutional, and environmental factors associated with smallholder farmers' exit from agriculture, rather than to establish causal relationships. Specifically, it addresses three core research questions: (i) Which structural, financial, and behavioral factors are most strongly associated with smallholder farmers' exit? (ii) How do these associations vary across subgroups characterized by financial inclusion, religious constraints, and regional vulnerability? and (iii) What policy-relevant insights can be drawn to support inclusive and faith-sensitive rural finance reform? This framing ensures coherence between theoretical foundations, methodological design, and interpretive analysis.

This study emerges as a response to these questions, grounded in direct engagement with farming communities across Pakistan. Through surveys, field observations, and previous studies (Yi et al. 2021). Evidence reveals that smallholder farmers need financing to purchase essential agricultural inputs such as seeds, fertilizer, and machinery. Yet, many remain excluded from formal credit systems, even when they possess land as collateral. Under Islamic law, interest (*Riba*) is strictly prohibited, making conventional bank loans religiously impermissible (*Haram*) (Siddique & Siddique 2025). Even "interest-free" government loans are often rejected, as some scholars argue they are not *halal* if the interest is subsidized or repaid indirectly (Kuran 1995). Consequently, many smallholder farmers turn to informal lenders, local traders, or input dealers who impose significant markups. For instance, a \$100 input might cost \$120 on a six-month installment, representing a 20% premium. Although a \$100 bank loan repaid at \$110 may seem cheaper, the presence of interest makes it religiously unacceptable for devout Muslim farmers. Islamic financial instruments such as *Murabaha* offer Shariah-compliant alternatives (Miah & Suzuki 2020), but remain largely inaccessible due to bureaucratic obstacles, limited awareness, and institutional inefficiencies. As a result, smallholder farmers remain trapped in informal and often exploitative credit cycles that deepen poverty and hinder rural economic progress. While the religious rationale is specific to Pakistan's context, the underlying financing frictions resemble rural credit constraints observed in many developing regions.

Further, the financial struggles of smallholder farmers are compounded by the growing dominance of institutional and corporate farming, particularly by state-linked entities that have recently begun cultivating large areas of land using advanced technologies (Broadcasting, 2023; Tanwar 2024). These influential actors benefit from political patronage, advanced mechanization, institutional backing, and privileged access to government procurement centers (Binswanger & Deininger 1997). Their large-scale operations reduce reliance on rural labor, thereby displacing traditional smallholder farmers

and reducing employment opportunities. Government procurement programs, originally designed to guarantee fair prices, are often monopolized by politically connected landlords, forcing smallholder farmers to sell their crops to intermediaries at prices below production costs (Gupta et al. 2017). Poor infrastructure, rising fuel prices, and distant markets further erode profitability for small producers. While powerful entities navigate the system with ease, leveraging subsidies, direct contracts, and access to storage, many smallholder farmers fall deeper into debt, abandon farming, or migrate abroad for low-wage employment, often leaving dependents behind. When remittances fail, entire households risk collapsing into poverty. This rural exodus threatens both national food sovereignty and social stability, underscoring the urgent need for systemic reforms to protect smallholder farmers and promote equitable agricultural development.

While previous research has contributed essential insights into agricultural finance in Pakistan and other developing regions, such as disparities in credit access (F. U. Khan et al. 2024a, b) financial constraints frameworks (Hussain & Thapa 2012), and the role of interest-free credit in smallholder market participation (Sher et al. 2024), many studies have overlooked deeper structural and behavioral drivers. Specifically, existing literature rarely addresses institutional monopolies, input price asymmetries, or the religious and sociopolitical barriers that shape farmer behavior. Moreover, studies tend to focus on active farmers, neglecting those who have already exited agriculture, leaving a gap in understanding farmers' micro-level motivations and systemic challenges. Our study addresses these gaps by using a novel dataset to describe patterns linked to exit without claiming causal effects and applying machine learning techniques, including CatBoost, which has consistently demonstrated superior performance in terms of accuracy, interpretability, and generalizability (Fan et al. 2024; Kumar et al. 2025; Mahesh & Soundrapandiyam 2024). Its native handling of categorical data, ordered boosting, and symmetric trees enhances convergence and reduces overfitting (Bentéjac et al. 2021). In contrast, models like XGBoost and LightGBM require substantial tuning, while logistic regression and SVM cannot capture complex, nonlinear relationships. Deep learning approaches are excluded due to interpretability issues and limited data size (Sharma et al. 2024). We also incorporate natural hazard exposure to floods and pests to reflect environmental pressures affecting farm viability. By combining CatBoost via recursive feature elimination and cross-validation (RFECV) with SHapley Additive Explanations (SHAP) for interpretability, logistic regression for marginal effect estimation, and subgroup exit analysis for behavioral context, this study connects data-driven diagnostics with human-centered policy insights.

To diagnose the complex associations (drivers) of agricultural exit, we adopt an integrated computational framework rooted in artificial intelligence and data science. Initially, we apply CatBoost with RFECV to select the top-ranked features from the overall feature set. RFECV iteratively ranks and removes less informative or redundant features via cross-validation, thereby reducing overfitting while improving model accuracy. The selected features are then used to train and evaluate eight machine learning models, including categorical boosting (CatBoost), extreme gradient boosting (XGBoost), random forest (RF), gradient boosting (GB), support vector machine (SVM), logistic regression (LR), light gradient boosting machine (LightGBM), and decision tree (DT). To move beyond farmers' exit prediction and ensure model transparency, we apply SHAP to

interpret the influence and direction of each feature on exit risk at both global and individual levels. To improve policy relevance, we use LR to estimate marginal effects, offering interpretable probabilities rather than log-odds. Finally, subgroup exit rate analysis provides behavioral context, reinforcing the real-world applicability of our findings. This integrated methodology prioritizes interpretability and associative insight, supporting evidence-based, human-centered policy development.

The study offers three contributions. First, it integrates machine learning and econometric approaches to provide a comprehensive diagnostic analysis of the factors associated with farmer exit behavior. Second, it introduces a novel dataset including both current and former farmers, offering deeper insight into overlooked behavioral and institutional factors. Third, it identifies informal credit burdens, religious finance barriers, and procurement monopolies as critical drivers and recommends actionable and explainable policies to promote inclusive rural finance, reverse displacement, and align agricultural systems with the Sustainable Development Goals.

To the best of our knowledge, this is the first study to combine machine learning with an econometric framework to examine both structural and behavioral drivers of smallholder farmers' exit from agriculture in financially constrained and institutionally complex rural settings, especially where religious restrictions shape financial choices. Our approach is adaptable and can be re-estimated with local data in other regions, even where credit contracts and market structures differ from those in Pakistan.

The remainder of this paper is organized as follows. "[Literature review](#)" section provides a brief literature review. "[Data and methodology framework](#)" section provides the details of the dataset and methodology. "[Results and discussion](#)" section presents the results and analysis. "[Conclusion and policy recommendation](#)" section concludes the study with policy recommendations and future research directions.

Literature review

Theoretical review

The rising exit of smallholder farmers despite overall agricultural growth presents a critical paradox that requires interdisciplinary investigation. This study draws on perspectives from agrarian economics, behavioral finance, institutional theory, and rural development to explore the structural, financial, and socio-behavioral forces behind this phenomenon (Streletskaia et al. 2020). Central to this inquiry are issues of economic exclusion, informal debt dependency, and systemic inequality in rural regions of developing countries (Dwyer 2018). The theoretical framework integrates Push–Pull Migration Theory (Lee 1966), Structural Constraints Theory (Sen 1982), and the Marxian perspective on land and power relations in rural economies (Marx 2024). To explain the multi-dimensional forces shaping farmers' exit decisions, unlike previous studies that treat these theories in isolation, this research synthesizes them into a single analytical lens that links behavioral motivations (push–pull), institutional barriers (structural constraints), and class, power asymmetries (Marxian dispossession) as interacting determinants of agricultural exit. This review deepens understanding of farmer exit through theoretical and empirical insights, especially in rural developing regions.

Originating from migration studies, the Push–Pull Theory posits that individuals relocate due to adverse “push” factors in their origin and attractive “pull” factors

in destinations (Lee 1966). In the agricultural sector, push factors such as debt burdens, exploitative credit systems, and market inefficiencies are associated with exit from farming. At the same time, pull factors like non-farm income or urban employment encourage them to seek alternative livelihoods (F. U. Khan et al. 2024a, b). In developing nations, the dominance of informal credit (push) and remittance-based income (pull) aligns with this framework. Farmers trapped in debt cycles face diminished agricultural returns, while overseas employment opportunities incentivize exit, reflecting the theory's dual mechanisms (Nidhiprabha 2019). However, the theory's linearity may underestimate structural barriers (e.g., religious financial constraints) that complicate exit decisions, necessitating integration with institutional perspectives.

Structural Constraints Theory suggests that poverty and livelihood insecurity stem not only from resource scarcity but also from systemic barriers that restrict access to land, markets, credit, and substructure (Binswanger & Deininger 1997; Sen 1982). In developing countries, particularly Pakistan, these constraints include supply chains monopolized by politically connected landlords and state-linked agribusinesses (Shoukat & Gomez 2020), while inadequate rural infrastructure and market distance increase transaction costs for remote farmers (Matungul et al. 2001). Furthermore, Islamic prohibitions on interest (Riba) and limited awareness of alternative financing mechanisms, such as Murabaha, exclude many smallholder farmers from formal credit systems, forcing them to rely on exploitative informal lenders (Abedifar 2019). These interlocking barriers reinforce cycles of poverty and exclusion, existing as a structural inevitability rather than personal choice, thereby extending the push–pull model into the realm of institutional inequality.

Applying Marxian theory helps reveal how institutional capture, financial exclusion, and socio-religious constraints collectively drive smallholder exit, not by choice, but through structural coercion and dispossession. Marxian Dispossession Theory defines how capitalist systems expand by appropriating land, labor, and resources from marginalized groups for elite benefit (Brooks 2003; Harvey 2017; Perreault 2013). In Pakistan, state-linked agribusinesses and corporate actors monopolize land, subsidies, and procurement, displacing smallholders (Shoukat & Gomez 2020). Although Islamic finance offers options such as Murabaha, many farmers lack access to or awareness of them. Consequently, informal lenders exploit religious prohibitions on interest by providing costly, unregulated credit without oversight (Abedifar 2019; Kuran 1995). This dynamic presents agricultural exit as structural coercion shaped by institutional power and financial exclusion. Previous studies often overlook how religious financial barriers and elite control reinforce dispossession.

Together, these three theoretical pillars create a cohesive framework: Push–Pull Theory captures the micro-level motivations behind farmers' decisions, Structural Constraints Theory highlights institutional and infrastructural barriers, and Marxian Dispossession Theory reveals the broader political-economic forces that sustain inequality. This integrated perspective deepens the study's explanatory power and enhances its policy relevance. Conceptually, it connects “push” factors (debt, low profit, market distance), “structural” barriers (credit exclusion, monopolized procurement, religious financing limits), and “dispossession” forces (land capture, elite

control) to farmers' exit outcomes, guiding the selection of variables such as financial ratios, institutional access indicators, and regional disparities that empirically ground these theoretical insights.

Empirical review

Research on agricultural exit has identified diverse structural, economic, and behavioral drivers across regions, yet significant gaps persist. Remarkably underexplored is how limited awareness and access to religious financial norms, coupled with institutional asymmetries, shape farmer decisions in developing economies. Studies from high-income countries offer foundational insights but exhibit limitations. A county-level research in the US analysis demonstrated that off-farm employment and federal support influence exit rates, though effects varied spatially and lacked individual-level behavioral depth (Goetz & Debertin 2001). Similarly, Zhan et al. (2012) identified landholding size, farming income share, and labor availability as critical predictors of exit from grain production in China. However, their focus on scale and grain-specific economics overlooks the pervasive financial access barriers and institutional asymmetries characteristic of farmers in developing countries like Pakistan. Consequently, existing research inadequately addresses the religious, financial, and credit system constraints influencing farmer behavior where formal loan access is restricted.

Closer to Pakistan, studies of smallholder farmers also reveal critical omissions. Ahmad et al. (2020) utilized LR to identify crop input credit and landholding size as crucial drivers of exit in Sindh. While methodologically interpretable, their model neglected nonlinear patterns, variable interactions, and, significantly, the role of religious finance behaviors and access to market infrastructure. Sher et al. (2024) examined interest-free credit schemes but did not address the underlying behavioral reluctance linked to religious interpretations of financial instruments. Furthermore, a pervasive methodological flaw across many studies is the exclusion of farmers who have already exited agriculture. This introduces survivorship bias, omitting the experiences of the most vulnerable populations displaced by structural constraints and potentially underestimating the coercive influence of market forces, informal debt cycles, and religious exclusion on persistence decisions. Our study rectifies this by sampling both current and former farmers, enabling a more comprehensive analysis of exit determinants using advanced techniques.

Concurrently, machine learning shows promise in agricultural applications, though its use for socioeconomic exit analysis remains nascent. Gradient-boosting algorithms such as CatBoost have been shown to outperform traditional models in crop yield prediction, offering high accuracy and interpretability using biophysical data (Badshah et al. 2024; Mahesh & Soundrapandiyani 2024). However, these studies did not apply these advanced methods to socioeconomic or institutional factors affecting farmer livelihoods. Similarly, Kumar et al. (2023) employed XGBoost for rainfall prediction, but not for behavioral outcomes like exit. While machine learning models have been used to examine credit access factors, such as Dang et al. (2020) comparison of RF and multinomial logit for formal vs. informal loan participation in Vietnam, highlighting collateral and financial literacy, these studies rarely extend to exit behavior or structural exclusion rooted in religious norms. A study by Connor et al. (2025) in formal institutional settings links minority farmer delinquency to their underrepresentation in agricultural institutions,

but often overlooks contexts influenced by religious financial norms. Similarly, Katchova & Ahearn (2017) US cohort analysis, estimating an 8.5% annual exit rate linked to operational duration, captures temporal dynamics but neglects micro-level behavioral factors and cultural-institutional dimensions, especially religious financing frameworks central to Muslim-majority rural areas.

Our work bridges these theoretical and empirical gaps by applying CatBoost with RFECV, and SHAP analysis to unpack the direction, magnitude, and interaction of socio-economic, religious financial, and institutional variables influencing smallholder exit, a novel application of explainable AI in development economics, while integrating LR with marginal effects and exit rate quantification for both analytical power and inferential depth. This integrated approach reveals how Islamic financial exclusion, informal debt, and market constraints jointly drive involuntary exit in Pakistan (linking behavioral, structural, and power-based factors), offering contextually grounded insights for academia and rural policy.

Data and methodology framework

Data collection and processing

This study employed a structured questionnaire-based survey to investigate the factors associated with the exit of smallholder farmers in Pakistan. To collect data, field teams conducted face-to-face interviews in 30 districts between May 2024 and February 2025. The study is cross-sectional, and each household was interviewed once. The extended field period reflects staggered deployment across agroecological zones and the difficulty of tracing farmers who had exited, were mobile, or had migrated. Sampling was purposive to capture variation across provinces, market access, farm sizes, and factors associated with exit from farming. It is not a nationally representative probability sample and is therefore not used to estimate national prevalence. This design aligns with the study's exploratory and diagnostic aims, which prioritize identifying associative patterns over statistical generalization. We defined an exited farmer as a household reporting no cultivation in the current season and no intention to cultivate in the next two seasons. To minimize survivorship bias, we used community tracing, snowball referrals, key informant lists, and phone follow-ups to locate households that had exited farming.

Recognizing that many experienced farmers in rural areas lack formal education, the questionnaire was primarily administered by trained research assistants fluent in both local dialects and official languages, a method consistent with established field studies (Peralta et al. 2020). They followed a bilingual script with neutral wording and read questions aloud when needed. To minimize recall error, time-bounded items were anchored to the last harvest and the most recent complete season. Interview dates and location identifiers (district and tehsil) were recorded to enable temporal and spatial checks. All responses were entered on paper forms or tablets using standardized skip patterns. Among the literate farmers, 117 completed the questionnaire independently (104 current and 13 former).

Initially, 519 responses were collected across 20 features and one binary target variable. After quality control, 500 complete and consistent observations were retained, comprising 421 current and 79 exited farmers. Nineteen records were removed due to missing values or unresolved inconsistencies. Paper forms were double-entered into a secure

database, discrepancies were reconciled, and range and logical checks were applied. Ambiguous entries were clarified by brief call-backs where possible. The final dataset reflects the natural prevalence of agricultural exit in the field and provides a robust foundation for empirical analysis. A sample copy of the survey is provided in Appendix A.

While this dataset provides unique micro-level insights into smallholder behavior, three limitations merit attention. First, purposive sampling introduces potential selection bias, as exiters who migrated or dissolved their holdings entirely may remain underrepresented. Second, the cross-sectional design captures associations rather than causation, limiting temporal inference. Third, SMOTE balancing, while methodologically valid, may slightly alter the underlying feature distribution. These limitations are transparently acknowledged, and results are interpreted as diagnostic rather than predictive of national-level trends.

Preprocessing data

Before proceeding with model development, we undertook a rigorous data preprocessing phase to ensure the dataset's accuracy, consistency, and analytical strength. This essential step laid the foundation for meaningful insights, much like preparing the soil before sowing seeds. All categorical variables were numerically encoded; missing values were carefully handled to prevent bias; and all features were normalized to a 0–1 scale for consistency. After cleaning and encoding, the dataset was divided into input features and a target variable representing farmer exit status. The cleaned dataset was then randomly split into training (80%) and testing (20%) subsets, preserving the original distribution of exit and non-exit cases. This balanced split enhanced model reliability and generalizability, ensuring that exit predictions were both robust and reproducible.

Categorical encoding

Categorical features were converted into numerical form using one-hot encoding to prepare them for analysis. This process allowed the model to interpret key variables such as land size, input methods, and income sources. Since machine learning algorithms require numerical input, encoding was essential. One-hot encoding fairly represented each category without introducing artificial order, ensuring the data remained both accurate and suitable for modeling (Lu et al. 2024).

Feature normalization

To mitigate biases arising from disparate measurement scales across features, all our quantitative indicators were normalized to a [0,1] range using the approach by (Rozo et al. 2023). This process ensures equitable algorithmic treatment of variables, preventing dominance by high-magnitude features and enabling robust comparisons during model training. Without normalization, machine learning models risk misinterpreting relationships, prioritizing numerical extremes over contextually meaningful patterns. Harmonized scales enhance fairness and interpretability, grounding insights in actual dynamics.

Class imbalance mitigation

We applied the Synthetic Minority Over-sampling Technique (SMOTE) to address the class imbalance in dataset 421/79. SMOTE generates synthetic instances of the minority class (farmer exits) by interpolating within the existing feature space, effectively balancing the dataset. This approach enhances model performance by improving precision and recall while mitigating biased exit predictions. Unlike under-sampling, which risks information loss, or random over-sampling, which may lead to overfitting, SMOTE preserves our data integrity by generating realistic and meaningful instances. Recent studies have validated its effectiveness in handling class imbalance and improving predictive accuracy (Azhar et al. 2022; Sun et al. 2018). In this study, integrated with advanced machine learning models, SMOTE was chosen over alternative techniques for its ability to balance data without compromising quality.

Methodology

To identify the drivers of smallholder farmers' exit in Pakistan, we developed a four-stage framework combining machine learning, interpretability techniques, classical econometrics, and subgroup analysis. First, CatBoost with RFECV selected the most predictive features. Second, SHAP values quantified feature importance and directional effects. Third, LR with marginal effects estimated the conditional probability of exit, controlling for key covariates. Fourth, subgroup analysis compared exit rates across credit-dependent and land-size-defined groups. This approach bridges data-driven discovery with theoretical grounding, linking behavioral (push–pull), institutional (structural constraints), and structural–political (Marxian) perspectives to translate statistical patterns into actionable socioeconomic insights. The complete workflow is depicted in Fig. 1.

Feature selection and modeling

As is typical in real-world datasets, not all collected variables contributed meaningfully to the predictive task. Several features showed low variance, high multicollinearity, or minimal explanatory power, which could increase the risk of overfitting and reduce model generalizability. To address this, feature selection was implemented to enhance model reliability, efficiency, and interpretability. We employed RFECV with CatBoost as the base estimator, a method particularly well suited for our structured data. Configured with $\text{step}=1$, $\text{cv}=5$, and $\text{scoring}=\text{"roc_auc"}$, the RFECV procedure iteratively removed

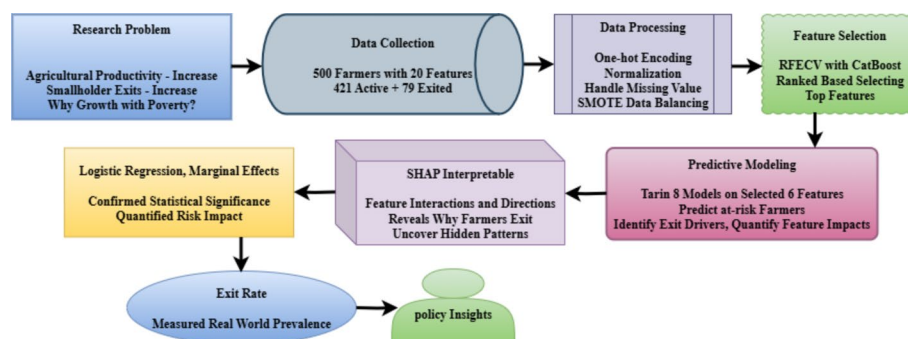


Fig. 1 Flowchart of the proposed framework

the least essential features while evaluating model performance at each step. This process resulted in a compact, high-impact subset of features that improved computational efficiency, reduced redundancy, and preserved the more informative features for subsequent modeling.

After feature selection, eight well-established machine learning models were trained and evaluated to investigate the factors influencing smallholder farmers' decisions to exit farming. This diverse set was chosen to ensure a balanced comparison across ensemble methods, tree-based learners, and linear classifiers. Particular attention was given to CatBoost, a gradient boosting algorithm developed by Prokhorenkova et al. (2018), due to its strong compatibility with the structure of our dataset. CatBoost provides several advantages for structured data, including ordered boosting to prevent target leakage, native support for categorical variables, and random permutation to reduce overfitting (Bentéjac et al. 2021). These characteristics made CatBoost especially suitable for our application, where accurate modeling and interpretability were both critical. Evaluating these models within a consistent framework allowed for a robust and generalizable algorithmic performance (Gao et al. 2023) while ensuring methodological rigor in identifying patterns associated with smallholder agricultural exit.

We performed hyperparameter tuning using RandomizedSearchCV to optimize the performance of each model. For CatBoost, key parameters such as depth, iterations, learning_rate, l2_leaf_reg, and border_count were fine-tuned to improve generalization and reduce prediction error. Similarly, appropriate parameter spaces were explored for the other models: max_depth, n_estimators, and gamma for XGBoost; C, kernel, and gamma for SVM; and num_leaves and min_child_samples for LightGBM. Once optimal configurations were identified, each model was trained on selected features. Model evaluation was conducted using multiple performance metrics, including accuracy, precision, recall, F1-score, geometric mean, and AUC-ROC. To determine the optimal classification threshold, we applied Youden's J statistic, which balances sensitivity and specificity, particularly important for imbalanced outcomes. To assess model stability and ensure generalizability, we used stratified k-fold cross-validation. All models were implemented using Python's scikit-learn and CatBoost libraries, ensuring reproducibility and methodological transparency. This comprehensive modeling pipeline enabled accurate, replicable analysis of smallholder exit behavior and supports evidence-based recommendations for policy targeting and risk mitigation.

SHAP for interpretability

To improve interpretability, we employed SHAP, a unified framework based on cooperative game theory (Lundberg & Lee 2017). While models such as CatBoost offer high predictive accuracy, their complexity can hinder transparency in high-stakes decisions, such as predicting agricultural exit. SHAP addresses this by quantifying each feature's contribution to individual predictions, offering both global and local interpretability (Lundberg et al. 2020).

Following model training and RFECV-based feature selection, we constructed refined datasets X_{train}^{SHAP} and X_{test}^{SHAP} retaining only the most informative features. A SHAP explainer was initialized using the trained CatBoost model and X_{train}^{SHAP} , allowing computation of SHAP values ϕ for each observation in X_{test}^{SHAP} .

At the core of SHAP is the decomposition of the model output $f(x)$, where $x \in R^n$ represents an input vector of n features. SHAP approximates the prediction as:

$$f(x) = \phi_0 + \sum_{i=1}^n \phi_i \quad (1)$$

where ϕ_0 is the expected model output, and ϕ_i quantifies the contribution of the i^{th} feature. The SHAP value ϕ_i is calculated as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{xi\}) - f(S)] \quad (2)$$

where N is the set of all features, and S is any subset excluding the feature i .

To quantify global importance, the mean absolute SHAP values were computed for each feature i overall test samples m :

$$Mean(|\phi_i|) = \frac{1}{m} \sum_{j=1}^m |\phi_i^{(j)}| \quad (3)$$

This aggregation reveals key features influencing exit decisions. Global and individual insights were visualized using summary, bar, and SHAP force plots. SHAP helped us explain why a particular farmer was likely to exit or remain in agriculture:

$$f(x^{(j)}) = \phi_0 + \sum_{i=1}^n \phi_i(j) \quad (4)$$

SHAP interpretability enhanced the transparency and applicability of our findings to policy.

Logistic regression and marginal effects analysis

To complement the machine learning results and deepen our understanding of feature influence, we applied LR using the same set of selected variables. LR is appropriate for binary outcomes and allows for direct interpretation of direction, magnitude, and statistical significance (Stoltzfus 2011).

Let $Y_i \in \{0, 1\}$ denote the binary response variable for the farmer i , where $Y_i = 1$ indicates an exit from agriculture. Let $X_i = (X_{i1}, X_{i2}, \dots, X_{ik})$ represent a vector of explanatory variables. The logistic model estimates the probability of exit as:

$$P(Y_i = 1|X_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_{i1} + \dots + \beta_K X_{ik})}} \quad (5)$$

where β_0 is the intercept, and $\beta_j = (j = 1, \dots, k)$ are coefficients associated with each feature. Parameters were estimated using maximum likelihood estimation (MLE). To enhance interpretability, we computed average marginal effects (AMEs), which translate log-odds into probability changes. For a continuous variable X_j , the marginal effect is:

$$\frac{\partial P(Y=1|X)}{\partial P} = \beta_j * P(Y = 1|X) * (1 - P(Y = 1|X)) \quad (6)$$

The average marginal effect over all n observations is:

$$AME_j = \frac{1}{n} \sum_{i=1}^n \beta_j * p_i * (1 - p_i) \quad (7)$$

where $p_i = P(Y = 1|X)$ For categorical features, marginal effects capture the discrete change in predicted probability when a category shifts from the baseline to an alternative level (Howell-Moroney, 2024). We used `get_margeff()` from `stats models` to estimate these effects, making the results easily interpretable in a policy context.

Exit rate

To further support and contextualize our findings, we conducted an exit rate analysis to quantify the share of farmers exiting agriculture across different subgroups. The exit rate was calculated as the proportion of farmers within each group who reported having exited, which aligns with approaches used in well-known studies (Katchova & Ahearn 2017).

$$Exit\ Rate_g = \frac{\sum_{i \in g} Y_i}{n_g} \quad (8)$$

where $Y_i = 1$ if farmer i exited agriculture, and n_g is the total number of farmers in group g . This nonparametric metric provides a direct and intuitive measure of exit risk without assuming a specific data distribution. We calculated exit rates as the groupwise mean of the binary outcome using the `pandas groupby(). mean()` function for efficient implementation. This subgroup-level analysis provided an additional lens for interpreting vulnerability and risk, complementing model-based predictions, SHAP explanations, and logistic inference.

Results and discussion

This study employed Python as the primary programming language for data processing, model training, and evaluation. All model development and testing were conducted in Visual Studio Code (VS Code), which provided a robust and efficient environment for debugging and iterative experimentation. The study investigated the socioeconomic and systemic factors driving smallholder farmers to exit agriculture in Pakistan, despite overall growth in agricultural production. Using advanced machine learning, econometric analysis, and interpretable AI techniques on the farmers' dataset, we identified key factors associated with farm exit decisions and uncovered critical insights into this paradox. All reported results reflect statistical associations rather than causal effects. The analysis compared models trained on the original imbalanced data and on balanced training folds generated by synthetic over-sampling. Reporting both sets of results demonstrates robustness and avoids overstated performance on minority outcomes.

Feature selection and model performance comparison

To ensure accuracy, generalizability, and practical relevance, we began by identifying the top-ranked features through RFECV and CatBoost. This approach was chosen for its ability to automatically and objectively evaluate feature importance during the model training process. RFECV helped reduce noise, multicollinearity, and overfitting, common challenges in high-dimensional socioeconomic data, by iteratively removing the least informative features while tracking cross-validated performance. As a result,

Table 1 Model performance for farmer exit prediction on imbalanced data

Model	Accuracy	AUC	Precision	Recall	F1-Score	G-mean
CatBoost	0.933	0.959	0.904	0.821	0.861	0.885
XGBoost	0.918	0.947	0.872	0.801	0.835	0.868
Random Forest	0.905	0.938	0.881	0.777	0.826	0.850
LightGBM	0.901	0.931	0.846	0.712	0.773	0.817
Gradient Boosting	0.898	0.930	0.803	0.710	0.754	0.814
Logistic Regression	0.857	0.891	0.753	0.688	0.719	0.782
SVM	0.855	0.901	0.755	0.656	0.702	0.765
Decision Tree	0.837	0.877	0.710	0.597	0.649	0.726

Table 1 presents the performance metrics of eight ML models for farmer exit prediction on imbalanced data.

Table 2 Model performance for farmer exit prediction trained on SMOTE-balanced data

Model	Accuracy	AUC	Precision	Recall	F1-Score	G-mean
CatBoost	0.910	0.966	0.832	0.903	0.866	0.907
XGBoost	0.901	0.951	0.802	0.871	0.835	0.889
Random Forest	0.895	0.941	0.801	0.872	0.835	0.886
LightGBM	0.878	0.941	0.797	0.852	0.824	0.867
Gradient Boosting	0.864	0.933	0.758	0.841	0.797	0.855
Logistic Regression	0.847	0.901	0.733	0.815	0.772	0.834
SVM	0.838	0.911	0.705	0.798	0.749	0.821
Decision Tree	0.797	0.879	0.665	0.787	0.721	0.793

Table 2 presents the performance metrics of eight ML models for farmer exit prediction on SMOTE-balanced data.

seven high-impact variables were retained: agricultural inputs purchasing method, debt burden, land size, non-farming income source, distance to the nearest market, livestock ownership, and exposure to natural hazards. Selection was performed within each cross-validation loop to prevent information leakage. These features formed the foundation of our subsequent models. Their selection reflected both statistical significance and theoretical grounding in rural economic behavior, aligning closely with credit dependency theory, Structural Constraints Theory, and resource-based perspectives on resilience.

Using the refined feature set, we evaluated eight machine learning models using both the original imbalanced dataset and a SMOTE-balanced version to ensure robust performance assessment. As shown in Tables 1 and 2, CatBoost consistently demonstrated superior performance across key metrics. Table 1 reports performance on the original imbalanced data. CatBoost achieved the best overall balance of metrics with an accuracy of 0.933, AUC-ROC of 0.959, precision of 0.904, recall of 0.821, F1 score of 0.861, and G-mean of 0.885. It effectively captures nuanced patterns within imbalanced datasets, avoids target leakage through ordered boosting, and handles missing values natively. Other models, such as XGBoost, LightGBM, and RF, performed competitively but with lower recall. LR, SVM, and DT underperformed due to overfitting and instability. These values replace earlier inflated headline figures and align with realistic performance for socioeconomic data.

To check sensitivity to class imbalance, we trained models on folds balanced with SMOTE and evaluated performance on held-out data without synthetic examples. Over-sampling was applied only within training folds to avoid leakage of synthetic points into

validation or test data. As shown in Table 2. This procedure raised recall for the exited class across models; for instance, CatBoost recall increased to 0.903, while accuracy decreased slightly to 0.910. This trade-off is expected because balancing emphasizes the minority class. For policy targeting, a higher recall reduces the risk of missing vulnerable households. Figure 2 clearly summarizes these results; the red and blue lines trace the performance trends across models for each metric under balanced and imbalanced training conditions, respectively. While the bars represent individual metric values, the lines highlight cross-model patterns, such as improved recall after data balancing.

We report multiple metrics because they capture different aspects of performance for imbalanced outcomes. AUC-ROC assesses ranking quality over all thresholds. Recall measures sensitivity for the exited class and is critical when the cost of missing an at-risk farmer is high. Precision indicates how many flagged cases are truly exited and is vital for program efficiency. The F1 score summarizes the balance between recall and precision. The G-mean combines sensitivity and specificity and is informative when class sizes are unequal.

This pipeline demonstrates the value of combining machine learning with thoughtful preprocessing and embedded feature selection for policy-relevant predictive analytics in agriculture, benefiting government agencies, financial institutions, and rural development planners.

Understanding exit risks through SHAP

Although RFECV provided a compact and high-performing feature set, it did not reveal how individual features contributed to specific predictions. To bridge this gap, we applied SHAP to the trained CatBoost model. SHAP values quantify the direction and magnitude of each feature's impact on prediction, enabling both global and local interpretability grounded in game theory.

Figures 3 and 4 show that the agricultural input purchasing method emerged as the most influential predictor of exit risk. Farmers relying entirely on credit-based input

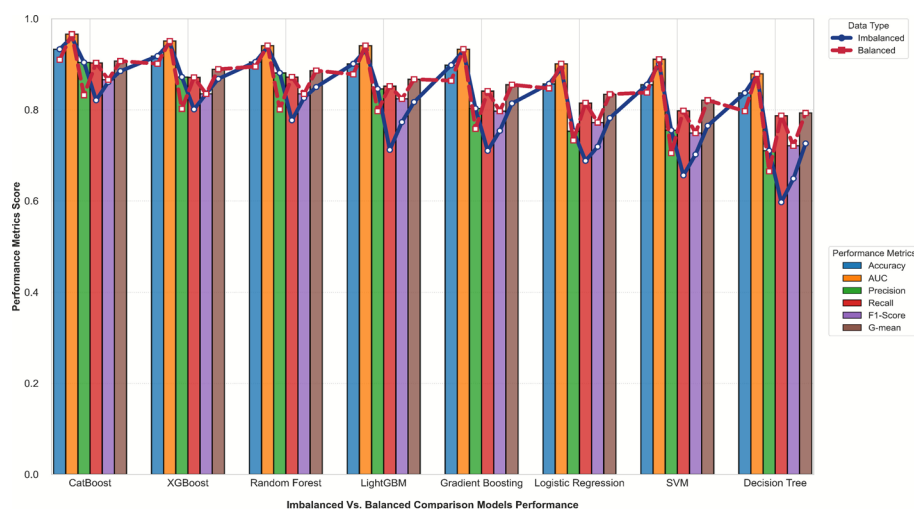


Fig. 2 Model performance comparison across metrics under imbalanced (blue line) and SMOTE-balanced (red line) training. Bars show individual metric values; lines trace trends across models

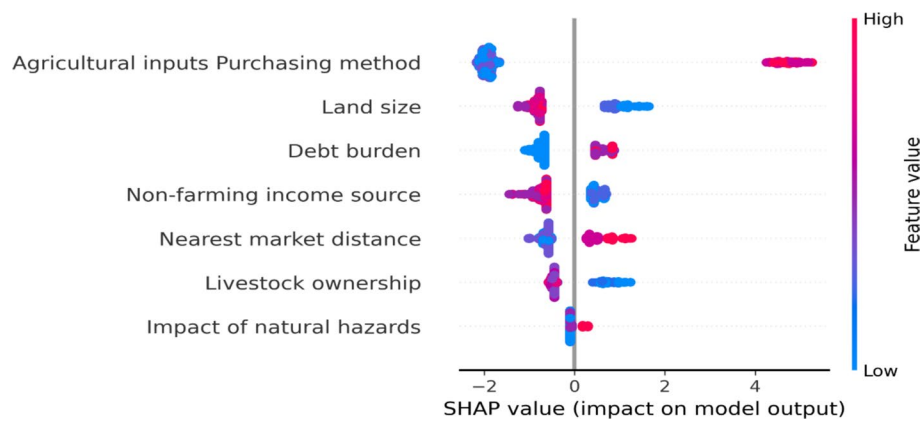


Fig. 3 SHAP summary plot illustrating directional feature impacts on farmer exit likelihood

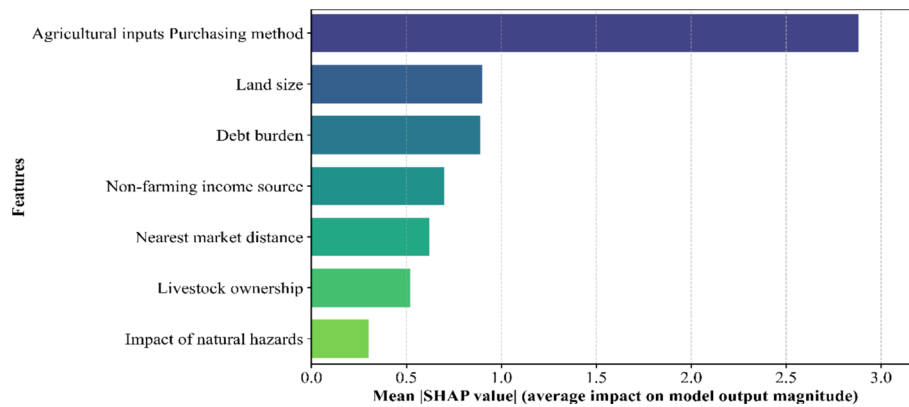


Fig. 4 SHAP bar plot showing average feature importance for predicting farmer exit risk

systems showed consistently positive SHAP values, indicating substantially increased exit probability. Economically, this means that dependency on informal or high-interest credit not only raises financial vulnerability but also reflects the “push” dynamic of the push–pull framework, where economic stress compels farmers to abandon cultivation. The result also supports Structural Constraints Theory, as it highlights how limited access to formal or Shariah-compliant financing creates systemic exit pressure. This finding highlights the critical role of financial arrangements in smallholder vulnerability, particularly in contexts where religious prohibitions or institutional barriers constrain formal credit access.

Debt burden constituted a highly important feature, with high debt levels strongly associated with increased exit risk. The positive relationship between debt and exit probability underscores the precarious position of smallholders facing financial stress, where production risks can quickly translate into existential threats to agricultural livelihoods.

Land size demonstrated a pronounced protective effect, with larger landholdings associated with reduced exit probability. This relationship shows nonlinear characteristics, with particularly sharp increases in vulnerability below specific threshold levels. Similarly, livestock ownership and non-farming income sources showed negative SHAP

values, indicating their role as protective factors that enhance resilience against economic shocks.

Market distance emerged as a significant structural barrier, with longer distances to markets strongly associated with increased exit probability. This relationship likely operates through multiple mechanisms, including reduced market access, higher transaction costs, and limited bargaining power for remote farmers. Exposure to natural hazards displayed modest positive contributions overall and was more pronounced in flood-prone localities, which broadens the analysis to environmental pressures without altering the ranking of the principal variables.

These variable-level relationships, particularly the roles of input financing, debt burden, asset buffers, and spatial isolation, align with and extend findings from prior studies on smallholder vulnerability and exit (Adams et al. 2021; Binam et al. 2017; M. Khan et al. 2024a, b; Viegas et al. 2023).

SHAP added unique value by uncovering nonlinear effects and interactions not captured by traditional methods. For instance, the combination of high debt and long market distance was associated with sharply increased exit risk, reinforcing the Structural Constraints Theory. It also revealed thresholds, such as a steep rise in exit risk below a specific land size, insights that are vital for setting policy triggers. Moreover, SHAP enabled case-level explanation: Why a specific farmer was classified as likely to exit, which is essential for frontline targeting.

Together, the balanced and imbalanced evaluations and the SHAP explanations support a cautious but practical interpretation. By converting complex model outputs into interpretable, visual narratives, SHAP transformed prediction into understanding. These insights provide policymakers with not only who is at risk, but why, an essential distinction when designing socially responsive interventions.

Logistic regression and marginal effects results

To enhance interpretability and provide formal statistical validation of our machine learning findings, we applied LR using the seven features selected through RFECV within the CatBoost modeling framework. While CatBoost combined with SHAP provided valuable insights into feature importance and directionality, these methods inherently lack inferential statistics (e.g., p -values, confidence intervals). LR addressed this limitation by quantifying the strength, direction, and statistical significance of associations between predictors and farmer exit outcomes. Six features were statistically significant at the 1% level, while natural hazards were significant at the 5% level. The model achieved a pseudo- R^2 of 0.405, which is consistent with structured behavioral data. We focus on marginal effects for interpretability and report coefficients for completeness. This statistical validation confirms the machine learning insights, enhancing their reliability for generalization and practical application in agricultural policy development.

The regression coefficients revealed key risk and protective factors, as shown in Table 3. The coefficients confirmed that purchasing inputs on credit ($\beta=6.242$), high debt burden ($\beta=4.216$), and market distance ($\beta=2.113$) were strongly positively associated factors, emphasizing the role of financial stress in pushing farmers out of agriculture. On the other hand, land size ($\beta=-1.782$), non-farming income ($\beta=-0.782$), and livestock ownership ($\beta=-0.735$) reduced exit probability (as protective factors).

Table 3 Logistic regression results

Features	Coef	Std. Err	z-value	p-value	95% CI
Agricultural inputs Purchasing method	6.242***	0.746	8.38	< 0.001	[4.780, 7.700]
Debt burden	4.216***	0.516	8.16	< 0.001	[3.200, 5.230]
Land size	− 1.782***	0.195	− 9.10	< 0.001	[− 2.170, − 1.400]
Non-farming income source	− 0.782***	0.117	− 6.69	< 0.001	[− 1.010, − 0.550]
Nearest market distance	2.113***	0.225	9.40	< 0.001	[1.670, 2.550]
Livestock ownership	− 0.735***	0.111	− 6.64	< 0.001	[− 0.950, − 0.520]
Impact of natural hazards (flood, pests)	0.140*	0.066	2.12	0.034	[0.011, 0.269]
Pseudo R ² = 0.405					

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4 Marginal effects results

Features	Marginal effects (dy/dx)	Std. Err	z-value	p-value	95% CI
Agricultural inputs Purchasing method	0.095***	0.021	4.53	< 0.001	[0.054, 0.137]
Debt burden	0.226***	0.022	10.20	< 0.001	[0.182, 0.269]
Land size	− 0.155***	0.011	− 14.69	< 0.001	[− 0.175, − 0.134]
Non-farming income source	− 0.092***	0.013	− 7.30	< 0.001	[− 0.117, − 0.067]
Nearest market distance	0.183***	0.012	14.91	< 0.001	[0.160, 0.208]
Livestock ownership	− 0.086***	0.012	− 7.31	< 0.001	[− 0.110, − 0.064]
Impact of natural hazards (flood, pests)	0.012*	0.006	2.00	0.045	[0.000, 0.024]

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Additionally, exposure to natural hazards (floods or pests) showed a smaller but statistically significant positive association ($\beta = 0.140$), particularly in flood-prone districts, a finding that extends the model's relevance to climate-vulnerable contexts without altering the hierarchy of core economic drivers, affirming that environmental shocks contribute to exit risk alongside structural factors. These findings align with the Push–Pull Theory of migration, where push factors such as debt and credit dependency drive farmers to leave agriculture, while pull factors like land, income diversification, and livestock ownership encourage continued engagement.

To further refine our understanding of how much each variable influences exit probability, we performed a marginal effects analysis, shown in Table 4. Purchasing inputs on credit is associated with a 9.5 percentage point higher predicted probability of exit compared with a cash purchase. High debt corresponds to a 22.6 percentage point higher probability. Each one-unit increase in land size is associated with a 15.5 percentage point lower probability, which is consistent with a buffering role of productive assets. A greater distance to market is associated with an 18.3 percentage point higher probability. The presence of non-farm income and livestock ownership is protective and corresponds to 9.2 and 8.6 percentage points lower probabilities, respectively. Natural hazards show a small but positive association and broaden the coverage of environmental pressures without changing the ordering of the principal variables.

These marginal effects provide a direct economic interpretation: Small, targeted interventions, such as reducing input credit dependency by 10%, capping debt–service ratios, or improving market access within 15 km, can yield measurable reductions

in exit risk. They offer a foundation for cost–benefit analysis of policy options, from input voucher programs to rural road investments.

Therefore, the LR results not only validate the machine learning outcomes but also translate them into an economically and theoretically coherent narrative of financial stress, asset inequality, and institutional exclusion. Together, LR and marginal effects analyses extend the predictive insights of machine learning by offering statistically grounded and interpretable findings that are both theoretically relevant and policy actionable.

Exit Rates results analysis

Moving beyond model-based estimates, we analyzed actual observed exit rates across farmer subgroups to ground our findings in empirical reality. This analysis reveals dramatic disparities in exit prevalence across different socioeconomic conditions, putting numerical values to the human experience of agricultural distress.

Farmers facing multiple structural disadvantages experience a near-certain exit, as detailed in Table 5. Those relying entirely on credit-based inputs faced a 72.6% exit rate, while those with high debt burdens reached 72.4% exit rates. The most vulnerable group, farmers with both credit dependence and high debt, experienced exit rates

Table 5 Farmer exit rate subgroup results

Features	Group (Code and Description)	Exit rate	Interpretation
Agricultural inputs purchasing method	1 = Full cash	2.0%	Negligible risk
	2 = More cash, some credit	2.0%	Negligible risk
	3 = More credit, some cash	67.6%	Critical risk
	4 = Full Credit	72.6%	Critical risk
Debt burden	0 = No debt	1.3%	Negligible risk
	1 = Low debt	19.9%	Moderate risk
	2 = High debt	72.4%	Critical risk
Land size	1 = Less than 5 acres	56.0%	High risk
	2 = 5 to 10 acres	32.4%	Moderate risk
	3 = 10 to 15 acres	5.5%	Low risk
	4 = 15 to 20 acres	5.4%	Low risk
	5 = More than 20 acres	3.3%	Negligible risk
Nearest market distance	1 = Less than 10 km	2.7%	Negligible risk
	2 = 10 to 20 km	4.7%	Negligible risk
	3 = 20 to 30 km	24.6%	Moderate risk
	4 = More than 30 km	55.2%	High risk
Non-farming income source	1 = Employment	15.2%	Moderate risk
	2 = Family members working abroad	57.0%	High risk
	3 = Own business in home country	3.3%	Negligible risk
	4 = Property rent	4.5%	Negligible risk
	5 = Others	3.4%	Negligible risk
Livestock ownership	0 = No livestock	55.5%	High risk
	1 = Horses and camels	10.8%	Low risk
	2 = Cows and buffalo	3.8%	Negligible risk
	3 = Goats and sheep	8.1%	Low risk
	4 = Chicken	15.2%	Moderate risk
Impact of natural hazards	5 = Others	18.8%	Moderate risk
	0 = No impact	2.9%	Negligible risk
	1 = Minor impact	13.4%	Low risk
	2 = Major impact	71.5%	Critical risk

Table 5 presents systemic inequalities in exit rates across risk categories, reflecting livelihood vulnerability.

exceeding 90%, highlighting how financial stressors compound to make continuation practically impossible.

Land inequality emerges as a fundamental determinant of resilience. Farmers with less than 5 acres faced a 56.0% exit rate, compared to near-zero exits (3.3%) among those owning more than 20 acres. This steep gradient demonstrates how land security provides crucial buffering against other shocks, while land poverty creates profound vulnerability. This aligns closely with Marxian Dispossession Theory, where land ownership determines control over production and survival; those with insufficient land become structurally coerced into exiting agriculture.

Spatial disadvantage similarly predicts exit patterns. Farmers located more than 30 km from markets experienced 55.2% exit rates, compared to only 2.7% among those within 10 km. This finding quantifies the exclusionary effects of infrastructure deficits and market inaccessibility that remote communities face.

Several protective factors showed striking effectiveness. Access to non-farm income, particularly through overseas remittances (57.0% exit rate) or local employment (15.2%), significantly enhanced resilience. Similarly, livestock ownership reduced exit rates substantially, from 55.5% among those without livestock to 3.8% for cattle/buffalo owners and 10.8% for camel/horse owners, suggesting different livestock types offer varying levels of economic security.

The integration of environmental shocks into our analysis, farmers reporting major hazard impacts faced 71.5% exit rates, compared to 2.9% among those reporting no impacts. However, this effect manifested primarily among already vulnerable groups, particularly smallholders with high debt and credit dependence, demonstrating how climate vulnerabilities intersect with structural disadvantages.

These exit rates highlight how vulnerability is not evenly distributed but structurally embedded. The patterns align with Marxian Dispossession Theory, showing how compounded disadvantages, credit exclusion, land inequality, and infrastructure deficits push out smallholders. Importantly, non-farming income and livestock emerged as powerful factors. Their presence significantly lowered exit rates, reinforcing the role of asset diversity and income buffering in rural resilience.

Together, these exit rate disparities confirm that farmer exits are not isolated economic choices but the cumulative outcome of interacting “push” pressures (debt, distance), structural barriers (credit exclusion, infrastructure), and dispossession forces (land inequality and institutional dominance).

Exit rate analysis brought the statistics to life. It reminded us that behind each prediction or percentage point lies a household navigating economic fragility, institutional failure, and limited opportunities. By connecting data to lived realities, this study provides not only a diagnostic tool but also a call to action for inclusive and preventive agricultural policy.

Conclusion and policy recommendation

This study addresses a critical and often overlooked paradox in Pakistan’s rural development: Despite rising national agricultural productivity, smallholder farmers are increasingly exiting agriculture under conditions of financial distress, institutional neglect, systemic exclusion, and facing deepening poverty. By combining machine learning,

interpretable AI, and classical econometrics, this study offers a novel, multi-dimensional understanding of farmer exit, rooted in behavioral, financial, and structural realities. All findings are presented as associations rather than causal effects, which is consistent with the cross-sectional design.

Our analytical framework integrated CatBoost with RFECV, SHAP interpretability, LR, and exit rate analysis. Feature selection using RFECV allowed us to isolate the highly influential attributes of exit behavior, filtering out redundant or low-variance features and strengthening model performance and interpretability. This step proved critical in simplifying a complex dataset without sacrificing accuracy, reducing noise, and enabling clearer insights. The CatBoost model, well suited for structured, categorical data, emerged as the most reliable classifier, reflecting its strength in handling small, imbalanced datasets while preserving nonlinear interactions. SHAP analysis then revealed how and why each selected feature influenced exit likelihood, while LR and marginal effects quantified these associations, offering grounded, statistically interpretable results.

Crucially, our analysis identified seven key drivers that are most strongly associated with agricultural exit: input purchasing method, debt burden, landholding size, non-farming income, market distance, livestock ownership, and impact of natural hazards. Together, these variables captured both economic vulnerability and structural disconnection. For example, full reliance on credit and heavy debt loads substantially increased exit probability, while larger land size, non-farming income, and livestock assets served as protective buffers. Additionally, natural hazards show a positive association with exit and broaden the coverage of environmental pressures. These findings confirm that farmer exits are not random or purely individual choices, but rather the result of entrenched financial constraints, institutional asymmetries, and infrastructure gaps. Exit rate analysis further humanized these findings, showing that nearly all farmers relying on full credit had exited, while those with more land, income diversification, or livestock remained resilient. These results validate key theoretical frameworks, including Push–Pull Migration, Structural Constraints, and Marxian Dispossession theories. Our study contributes meaningfully to global development discourse by aligning with multiple Sustainable Development Goals: SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 8 (Decent Work and Economic Growth), SDG 10 (Reduced Inequalities), and SDG 12 (Responsible Consumption and Production). By reframing agricultural exit as a symptom of structural exclusion rather than individual inefficiency, this study advocates for faith-aligned, inclusive, and evidence-based policy reform to support smallholder resilience.

Policy recommendations

We propose a modular policy toolkit that connects each empirical association with a practical intervention and measurable indicator. These recommendations are diagnostic in nature and suitable for pilot implementation before broader scaling.

First, to prevent involuntary smallholder exits, especially among farmers with less than ten acres, we recommend establishing a Smallholder Security Contract (SSC) program. This mechanism would guarantee government-provided essential inputs at cost and fair-price procurement of outputs, serving as a financial safety net against exploitative intermediaries and informal lending practices. This directly corresponds to the empirical

evidence showing that debt burden and credit-dependent input purchases significantly increase exit risk.

Second, financial inclusion must be expanded through genuinely Shariah-compliant instruments (e.g., Murabaha, Mudaraba, Musharakah, Qard Hasan) supported by mobile banking and simplified documentation endorsed by local Islamic scholars. National restructuring of informal debt is also essential to restore liquidity and trust in formal channels. This aligns with our finding that informal credit dependence is the most influential predictor of exit, underscoring the need for targeted, faith-sensitive credit reform.

Third, input access should transition from credit-linked subsidies to digital or paper-based voucher systems. Furthermore, digitizing land records and supporting cooperative land consolidation enhance tenure security and production scale. Physical accessibility is equally vital; decentralized procurement centers within 15 km of farming clusters, using clear e-procurement systems with smallholder quotas, can improve fairness and efficiency. For digitally excluded farmers, voice-based advisory systems in local languages and community-nominated proxy agents can bridge communication gaps. This recommendation is supported by the strong association between market distance and exit probability revealed in the SHAP and LR analyses.

Finally, embedding predictive analytics such as CatBoost within agricultural extension services can provide low-cost early warning systems. These systems would enable timely interventions, including agricultural training, livestock support, and remittance-linked programs targeting women and youth. Collectively, these strategies shift policy from reactive aid to proactive protection, realigning rural development with principles of equity, sustainability, and human dignity.

Overall, these recommendations are empirically bounded, focusing on measurable and evidence-driven actions, such as improving market access, reforming credit systems, and managing debt burdens, rather than broad institutional restructuring. This ensures coherence between the study's analytical findings and its policy implications, while maintaining academic rigor and policy relevance.

Limitations and future directions

This study offers significant insights into the structural drivers of smallholder exit but also highlights areas for future research. The cross-sectional design precludes causal inference, necessitating longitudinal and panel studies to track dynamic exit behavior over time and establish causal pathways. While our multi-method approach strengthens robustness, potential endogeneity from omitted variables and survivorship bias, particularly among completely displaced farmers, calls for innovative tracking methods, including mobile surveys, satellite imagery, and migration cohorts.

Further research should also integrate richer qualitative dimensions, such as emotional, generational, and nuanced religious drivers, through mixed-methods designs to complement our quantitative findings. Moreover, this analytical framework could be extended to other Muslim majority or minority regions where religious finance principles shape agricultural vulnerability. Comparative studies across diverse legal, institutional, and cultural settings, both within and beyond Islamic contexts, would help distinguish universal mechanisms from context-specific manifestations of exit behavior.

Despite these limitations, our study provides a robust foundation for both policy formulation and further research into smallholder vulnerability and resilience.

Appendix A. Survey variables and response coding

1. Education Level: (1) No formal education, (2) Primary school, (3) Middle school, (4) High school, (5) College degree, (6) Bachelor's degree and above

2. Age of Household Head: (1) 15–25 years, (2) 26–35 years, (3) 36–45 years, (4) 46–55 years, (5) 56–65, (6) 66 and above

3. Marital Status: (1) Single, (2) Married, (3) Divorced/widowed/widower

4. Total Family Members: (1) Less than 5, (2) Between 5–10, (3) More than 10

5. Family Members in Labor Force: (1) Less than 4, (2) Between 4–7, (3) More than 7

6. Family Secondary Skill status to earn money: (0) No, (1) Small business skills (shop, trade, etc.), (2) Technical skills (electrician, carpentry, etc.), (3) Agricultural machinery Expertise, (4) Other

7. Farming Experience: (1) Less than 10 years, (2) 10–30 years, (3) 30–50 years, (4) More than 50 years

8. Land size = (1) Less than 5 acres, (2) 5–10 acres, (3) 10–15 acres, (4) 15–20 acres, (5) More than 20 acres

9. Agricultural Inputs Purchasing Method: (1) Cash, (2) More cash, some credit, (3) More credit, some cash, (4) Fully credit

10. Irrigation Method: (1) Groundwater (tube well, bore), (2) Mostly Groundwater with some canal water, (3) Mostly canal/government water, (4) Canal/government water with some groundwater, (5) Other/rain/drip

11. Crop Types Grown: (1) Wheat, (2) Sugarcane, (3) Cotton, (4) Maze and Fodder, (5) Fruits and vegetables, (6) Rice, (7) Others

12. Livestock Ownership: (0) No livestock, (1) Horses and camels, (2) Cows and buffaloes, (3) Goats and sheep, (4) Chicken, (5) Others

13. Market Access/Where products sold: (1) To middlemen, (2) Government storage facilities, (3) Private mills or stores, (4) Local market, (5) Others

14. Nearest Market Distance: (1) Less than 10 km, (2) 10–20 km, (3) 20–30 km, (4) More than 30 km

15. Non-Farming Income Source: (1) Employment, (2) Family members working abroad, (3) Own business in home country, (4) Property rent (5) Others

16. Receiving Government Subsidies: (0) No, (1) Yes

17. Debt Burden: (0) No debt, (1) low debt, (2) High debt

18. Awareness of Government Supporting Programs: (0) Not aware of any programs, (1) Aware but not benefitting, (2) Aware and benefitting

19. Impact of Natural Hazards (floods and pests): (0) No impact, (1) Minor impact, (2) Major impact

20. Internet Usage: (0) No internet usage, (1) Only for social media, (2) For general news, (3) For agricultural-related information, (4) for all, (5) Others

Target Variable. Farmer exits or not: (0) No, (1) Yes

This table presents the structured variables from the household farmer survey. All variables were numerically coded to support statistical and machine learning analyses. The survey covers demographic, economic, and behavioral indicators, with a binary target variable for farmer exit. This transformation ensured consistent formatting and efficient, reproducible data processing.

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Author contributions

Author Contribution statement: Ghazi Abbas conceptualized and designed the manuscript, developed the methodology, and conducted the data analysis. Prof. Chi Guotai guided research methodologies and overall study direction. Prof. Zhou Ying, Qamar Abbas contributed to data collection and supported the formal analysis of the manuscript, and

Professor Khalil El Hindi was instrumental in redesigning the analytical framework, interpreting new computational results, and securing the vital financial support. All authors reviewed and approved the final version of the manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request, subject to ethical and confidentiality restrictions.

Declarations

Ethical approval and consent to participation

This article does not contain any studies with human participants performed by any of the authors.

Consent to publication

This article does not contain any studies with human participants performed by any of the authors.

Competing interests

The authors declare no competing interests.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Declaration of generative AI in scientific writing

During the preparation of this work, the author(s) used ChatGPT to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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References

- Abedifar P (2019) Lending, the poor, and Islamic scripture: Islamic finance versus welfare Islam. *J Am Acad Relig* 87(2):460–482
- Adams F, Ohene-Yankyera K, Aidoo R, Wongnaa CA (2021) Economic benefits of livestock management in Ghana. *Agric Food Econ* 9:1–17
- Ahmad MI, Oxley L, Ma H (2020) What makes farmers exit farming: a case study of Sindh Province, Pakistan. *Sustainability* 12(8):3160
- Ali S, Mueed A, Jahangir M, Sammi S, Zakki SA, Amin A, Anwar K, Ayoub A, Li P, Shehzad F (2024) Evolution of olive farming, industry, and usage in Pakistan: a comprehensive review. *J Agric Food Res* 16:101091
- Azhar NA, Pozi MSM, Din AM, Jatowt A (2022) An investigation of smote based methods for imbalanced datasets with data complexity analysis. *IEEE Trans Knowl Data Eng* 35(7):6651–6672
- Badshah A, Alkazemi BY, Din F, Zamli KZ, Haris M (2024) Crop classification and yield prediction using robust machine learning models for agricultural sustainability. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2024.3486653>
- Bentéjac C, Csörgő A, Martínez-Muñoz G (2021) A comparative analysis of gradient boosting algorithms. *Artif Intell Rev* 54:1937–1967
- Binam JN, Place F, Djalal AA, Kalinganire A (2017) Effects of local institutions on the adoption of agroforestry innovations: evidence of farmer managed natural regeneration and its implications for rural livelihoods in the Sahel. *Agric Food Econ* 5(1):2
- Binswanger HP, Deininger K (1997) Explaining agricultural and agrarian policies in developing countries. *J Econ Lit* 35(4):1958–2005
- Broadcasting Mola (2023) Federal Govt, Pakistan Army develop LIMS to modernize agriculture sector. Government of Pakistan. <https://www.moib.gov.pk/News/54443>
- Brooks RE (2003) The new imperialism: violence, norms, and the “rule of law.” *Mich Law Rev* 101(7):2275–2340
- Connor L, Ahrendsen BL, Dodson CB, Moss L (2025) Minority farmer and rancher delinquency and local cultural climate: evidence from Farm Service Agency’s county office committees. *Am J Agric Econ*. <https://doi.org/10.1111/ajae.12546>
- Dang HD, Dam AHT, Pham TT, Nguyen TMT (2020) Determinants of credit demand of farmers in Lam Dong, Vietnam: a comparison of machine learning and multinomial logit. *Agric Financ Rev* 80(2):255–274
- Dwyer RE (2018) Credit, debt, and inequality. *Annu Rev Sociol* 44(1):237–261
- Fan Z, Gou J, Weng S (2024) A feature importance-based multi-layer catboost for student performance prediction. *IEEE Trans Knowl Data Eng*. <https://doi.org/10.1109/TKDE.2024.3393472>
- Finance Division GoP (2024) Highlights - Pakistan Economic Survey 2023–24. https://www.finance.gov.pk/survey/chapter_24/Highlights.pdf
- Gao Y, Wang L, Zhong G, Wang Y, Yang J (2023) Potential of remote sensing images for soil moisture retrieving using ensemble learning methods in vegetation-covered area. *IEEE J Select Topics Appl Earth Observ Remote Sensing* 16:8149–8165

- Goetz SJ, Debertin DL (2001) Why farmers quit: a county-level analysis. *Am J Agric Econ* 83(4):1010–1023
- Gupta S, Dawande M, Janakiraman G, Sarkar A (2017) Distressed selling by farmers: model, analysis, and use in policy-making. *Prod Oper Manag* 26(10):1803–1818
- Harvey D (2017) The new imperialism: accumulation by dispossession. In *Karl marx* (pp. 213–237). Routledge.
- Howell-Moroney M (2024) Inconvenient truths about logistic regression and the remedy of marginal effects. *Public Adm Rev* 84(6):1218–1236
- Hussain A, Thapa GB (2012) Smallholders' access to agricultural credit in Pakistan. *Food Secur* 4:73–85
- Katchova AL, Ahearn MC (2017) Farm entry and exit from US agriculture. *Agric Financ Rev* 77(1):50–63
- Khan FU, Nouman M, Negrut L, Abbas J, Cismas LM, Siddiqi MF (2024a) Constraints to agricultural finance in underdeveloped and developing countries: a systematic literature review. *Int J Agric Sustain* 22(1):2329388
- Khan M, Papadas D, Arnold L, Behrendt K (2024b) Sustainability challenges in the multi-tier crop agri-food sector: a systematic review. *Agric Food Econ* 12(1):25
- Kumar V, Kedam N, Sharma KV, Khedher KM, Alluqmani AE (2023) A comparison of machine learning models for predicting rainfall in urban metropolitan cities. *Sustainability* 15(18):13724
- Kumar V, Kedam N, Kisi O, Alsulamy S, Khedher KM, Salem MA (2025) A comparative study of machine learning models for daily and weekly rainfall forecasting. *Water Resour Manage* 39(1):271–290
- Kuran T (1995) Islamic economics and the Islamic subeconomy. *J Econ Perspect* 9(4):155–173
- Lee ES (1966) A theory of migration. *Demography* 3:47–57
- Lu Z, Li H, Wu J (2024) Exploring the impact of financial literacy on predicting credit default among farmers: an analysis using a hybrid machine learning model. *Borsa Istanbul Rev* 24(2):352–362
- Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, Katz R, Himmelfarb J, Bansal N, Lee S-I (2020) From local explanations to global understanding with explainable AI for trees. *Nat Mach Intell* 2(1):56–67
- Lundberg SM, & Lee S-I (2017) A unified approach to interpreting model predictions. *Advances in neural information processing systems*
- Maresh P, Soundrapandian R (2024) Yield prediction for crops by gradient-based algorithms. *PLoS ONE* 19(8):e0291928
- Marx K (2024) *Capital: Critique of Political Economy*, vol 1. Princeton University Press
- Matungul PM, Lyne MC, Ortmann GF (2001) Transaction costs and crop marketing in the communal areas of Impendle and Swayimana. *KwaZulu-Natal Development Southern Africa* 18(3):347–363
- Miah MD, Suzuki Y (2020) Murabaha syndrome of Islamic banks: a paradox or product of the system? *J Islam Account Bus Res* 11(7):1363–1378
- Nidhiprabha B (2019) Commodity price cycles, the agricultural trap, and Thailand's incessant subsidies. *Asian Econ Pap* 18(2):49–69
- Oscar Barriga-Cabanillas SK, Moritz Meyer Muhammad Nasir Maria Qazi (2024) Poverty Projections for Pakistan (Poverty and Equity Global Department, Issue. <https://documents1.worldbank.org/curated/en/099459512302438604/pdf/IDU17bf933d2197611440918caf1f63cb8d5bfdc.pdf>
- Peralta A, Paniagua-Avila A, Fort M, Gulayin P, He J, Irazola V, Mansilla K, Hernández D, Ramírez-Zea M (2020) Translating instruments into multiple Mayan languages for a hypertension T4 study in Guatemala. *Eur J Public Health* 30(Supplement_5):ckaa166. 1160
- Perreault T (2013) Dispossession by accumulation? Mining, water and the nature of enclosure on the Bolivian Altiplano. *Antipode* 45(5):1050–1069
- Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, & Gulin A (2018) CatBoost: unbiased boosting with categorical features. *Advances in neural information processing systems*
- Rozo BJG, Crook J, Andreeva G (2023) The role of web browsing in credit risk prediction. *Decis Support Syst* 164:113879
- Sen A (1982) *Poverty and famines: an essay on entitlement and deprivation*. Oxford University Press
- Sharma A, Lysenko A, Jia S, Boroevich KA, Tsunoda T (2024) Advances in AI and machine learning for predictive medicine. *J Hum Genet* 69(10):487–497
- Sher A, Mazhar S, Qiu Y (2024) Toward sustainable agriculture: the impact of interest-free credit on marketing decisions and technological progress in Pakistan. *Sustain Dev* 32(1):608–623
- Shoukat A, Gomez ET (2020) Business groups, political connections, and regime change in Pakistan. *Asian Surv* 60(5):952–977
- Siddique Z, Siddique MA (2025) Riba and interest in Islamic jurisprudence: seeking the path to consensus. *Int J Ethics Syst* 41(2):543–561
- Stoltzfus JC (2011) Logistic regression: a brief primer. *Acad Emerg Med* 18(10):1099–1104
- Streletskaya NA, Bell SD, Kecinski M, Li T, Banerjee S, Palm-Forster LH, Pannell D (2020) Agricultural adoption and behavioral economics: bridging the gap. *Appl Econ Perspect Policy* 42(1):54–66
- Sun J, Lang J, Fujita H, Li H (2018) Imbalanced enterprise credit evaluation with DTE-SBD: decision tree ensemble based on SMOTE and bagging with differentiated sampling rates. *Inf Sci* 425:76–91
- Tanwar S (2024) Fighting to farming: Pakistan military's relentless quest for relevance. Observer Research Foundation. <https://www.orfonline.org/expert-speak/fighting-to-farming-pakistan-military-s-relentless-quest-for-relevance>
- Viegas M, Wolf J, Cordovil F (2023) Assessment of inequality in the Common Agricultural Policy in Portugal. *Agric Food Econ* 11(1):13
- Yi Z, Wang Y, Chen YJ (2021) Financing an agricultural supply chain with a capital-constrained smallholder farmer in developing economies. *Prod Oper Manag* 30(7):2102–2121
- Zhan JT, Wu YR, Zhang XH, Zhou ZY (2012) Why do farmers quit from grain production in China? Causes and implications. *China Agric Econ Rev* 4(3):342–362

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